

Train Smarter, Not More: Key Workouts

It's not that athletes were training incorrectly at the beginning of ultrarunning; it's that there wasn't really any specific training going on at all. Fit and very hardy individuals took on outlandish challenges and managed to finish them. More people became interested in these modern-day feats of endurance, and events became more organized and recognized. Through a lot of hard work and dedication from men and women we now know as icons of the sport, the events grew, course records were established, and a little bit of money came into the sport in the form of sponsorship and prize purses. However, as the winning times in the big ultramarathons started dropping, the sport developed beyond the point where "go run more" was a viable training strategy.

Ultrarunning isn't the only sport to experience this evolution. In fact, it's an evolution common to many sports. Triathlon, for instance, originated in 1977 in Mission Bay, California and in the early years the sport's top athletes were the ones who could endure the greatest workload. The sports science that had previously shaped the accepted practices in triathlon's component disciplines—swimming, cycling, and running—didn't catch up to the unique demands of triathlon until several years later. Training techniques, equipment, technology, and nutritional strategies all evolved—and continue to evolve—and course records continue to be broken.

One of the milestones for any sport occurs when training becomes organized and sport-specific, i.e., when "go run/train more" is replaced by specific kinds of activities that are separated by rest and organized into a schedule. Consider the concept of periodization, which has been around in various rudimentary forms for thousands of years. The modern and almost universally accepted version of periodization—systematically changing the focus and workload of training to maximize the positive impact of overload and recovery on training adaptations—was constructed by Tudor Bompa and other Eastern Bloc coaches in order to win Olympic medals in the 1950s, when the Olympics were as much about the battle of East vs West as they were about athletic achievement.

Before Bompa, German scientist Woldemar Gerschler took the relatively informal but highly effective training practices of Swedish running coaches and, in the 1930s, refined them into what you and I recognize today as interval training. At the time, the Swedes were using changes in terrain to interject periods of intensity and recovery into their longer runs. They referred to the practice as "fartlek" running, a term and technique which is still widely used. But Gerschler eliminated the unpredictability of fartlek training by adding structure—in the form of precise times, distances, and paces—so he could quantify both the work being done and the recovery being taken between efforts.

Neither Bompa nor Gerschler had you in mind when he was pushing the boundaries of sports science. Bompa had to earn Olympic medals to show the world the power of the Soviet system, and Gerschler was working to find a way to help his athletes, including eventual record-holder Roger Bannister, break the coveted four-minute barrier in the one-mile run. But the science they discovered changed the face of endurance training for athletes at all levels.

Interval training is effective for improving performance because it enables you to accumulate time at specific intensities, and it is time at intensity that is the most critical aspect to achieving positive adaptations. The interval workouts described in this chapter are designed to apply the principles of training—overload and recovery, progression, individuality, specificity, and a systematic approach (discussed in Chapter 3)—to individual training sessions. These workouts are in turn governed by five components of training: intensity, volume, frequency/repetition, environment, and stride rate. By manipulating these components, you can create a workout to target just about any physiological demand found in sport.

The Five Workout Components

When you get ready to head out for your training sessions, you can address the five principles of training using the following variables:

1. INTENSITY

2. VOLUME

3. FREQUENCY AND REPETITION

4. ENVIRONMENT (TERRAIN, SURFACE, AND AMOUNT OF VERTICAL RELIEF)

5. RUNNING CADENCE AND STRIDE RATE

You can change the goal of a workout by changing one of its components. The clearest example of this is found in changing the intensity. Two ten-minute intervals can target completely different aspects of your physiology if you simply change your pace. You can maintain a pace at the high end of your aerobic intensity to develop greater aerobic endurance, or you can run at your lactate threshold pace to increase your ability to produce and process lactate. You can also manipulate the environment for your workout (i.e., running uphill vs running on flat ground) to change its purpose.

INTENSITY

Intensity is a measure of how hard you are working, and the impact of a workout is directly related to the intensity at which you are working. Measuring intensity is notoriously difficult in running, particularly trail running. Unlike cycling, where athletes can utilize power meters to directly measure workload, or road running, where runners can generally use pace, trail runners can at best observe the body's response to workload in the form of RPE. Even gauging intensity by heart rate is problematic for trail runners, as explained in Chapter 6. New technologies may soon enable us to measure power or normalized pace in running, but until then the best measurement tool we have is RPE.

VOLUME

Volume is the total amount of exercise you're doing in a single workout, a week of training, a month, a year, or a career. The most important concept related to interval training volume is volume-at-intensity. One of the purposes of intervals is to break hard efforts into smaller bites so you can accumulate more time at a specific intensity than you would be able to sustain in one longer effort. For instance, if the goal of a workout is to spend sixty minutes at lactate threshold intensity, but currently you can sustain that pace for only forty minutes, you can organize the workout into six ten-minute intervals, separated by recovery periods. By the end of the workout, you have run at

your lactate threshold pace for sixty minutes and generated a training stimulus that will increase the amount of time you can stay at your threshold pace in a single effort. You can also schedule two or three of these workouts in a single week and accumulate up to 180 minutes of volume-at-intensity targeted at your lactate threshold pace.

To properly quantify volume, you have two choices: miles or time. Volume in an ultrarunning or trail running setting is best prescribed by time. This is because a five mile run can take you one, two, or even three hours depending on the terrain. When you view volume as a volume of time (rather than as a volume of distance), you can more accurately quantify training.

FREQUENCY AND REPETITION

Frequency is the number of times a workout is performed in a given training period, whereas repetition is the number of times an exercise is repeated in a single session. Frequency and repetition are used to ensure the quality of your training sessions. Intervals are effective only if they are performed at the prescribed intensity, volume, frequency, and repetition based on your current fitness. It is the correct combination of all of these variables that leads to the most effective training, not the dominance of any one variable over the others.

Let's say your lactate threshold pace is 9:00 min/mi, and you can sustain that pace for forty minutes. You might be able to run 8:30 pace for three minutes during a RunningInterval workout (defined later in this chapter), which would be closer to your pace at VO2 max. There's no point in trying to complete a twenty-minute RunningInterval because your pace would fall so dramatically after the first three to five minutes that the rest of the effort would no longer be VO2 max-type intensity. It would still feel ridiculously hard, but that effort would no longer be addressing the goal of a RunningInterval. In contrast, if you do seven three-minute RunningIntervals at 8:30 min/mi, separated by three minutes of easy running at a recovery pace, you'll accumulate twenty-one minutes at 8:30 pace.

Frequency gives you another way to accumulate workload, by repeating individual interval sessions during a given week, month, or even year. For instance, a week with two RunningInterval workouts like the one just described means forty-two minutes at 8:30 pace. The harder the intervals, the more recovery you need before you'll be ready to complete another high-quality training session. In any effective training program, workouts are spaced out to provide adequate recovery between sessions. I will discuss the relationship between interval workouts and recovery days later in this chapter.

ENVIRONMENT

You can use environmental factors to manipulate your workouts. In some cases, you can increase or decrease the workload of particular efforts, as with performing intervals while going uphill. As discussed previously, I prefer certain intervals, especially RunningIntervals, to be done uphill to maximize the aerobic benefit and to contribute to race-specific training, particularly if the goal race is hilly. Running your intervals uphill can also be useful as a hedge against injury and to overcome flagging motivation. Sometimes it can be difficult to push yourself through maximum-intensity intervals on flat ground, but a hill adds resistance and a visible challenge, which can be the little something extra you need to make your workout more effective.

Environment has a big impact on specificity in all sports. The surface you run on can make a difference in your pace and in the amount of stress you apply to your body. Even trail ultramarathons almost always have a mixture of surfaces, including trails of varying technical difficulty, dirt roads, and some pavement. Training is not just about developing cardiovascular fitness; it's also about preparing the musculature, joints, bones, and connective tissues for the challenges of competition.

You can use treadmills to your advantage in training. The suspension found in treadmills means that treadmill running typically has less impact on your feet and legs than running outdoors, and it even has less impact than

running on some trails. For athletes who have a history of foot or knee problems, running on a treadmill may enable you to complete more running with less pain. Running on the treadmill can also make interval training more convenient because you can program in the appropriate paces and durations and let the treadmill adjust your pace and hold it steady for the duration of the interval. This is often helpful for athletes who have trouble motivating themselves to maintain a fast-enough pace on their own for difficult intervals. In the end, you'll have to find that motivation to push yourself if you want to succeed on race day, but if you sometimes need the treadmill to provide that motivation, that's OK. Regardless of whether or not you will use a treadmill for interval workouts, EnduranceRuns, or RecoveryRuns, you can set the incline at various degrees depending on your goals.

RUNNING CADENCE AND STRIDE RATE

Decades ago, Cavanaugh and Williams (1982) showed that runners naturally gravitate toward the stride length that is most economical for them in terms of oxygen consumption. Numerous other investigations on running economy have reached similar conclusions. While your natural stride rate might change in the early phases of your running career, once you're a reasonably proficient runner, your biomechanics will settle into a sweet spot that is right for you.

For an ultrarunner, there are advantages to developing the ability to run at different stride lengths and rates. It may seem counterintuitive to purposely change your stride, given that you have gravitated toward the one that is optimal for you, but this makes sense in ultrarunning because changing your stride and foot-strike patterns changes the ways you are stressing your body. A faster stride rate with a shorter stride length, even if it is not your naturally chosen stride, changes the stresses on the muscles and ligaments in your legs. This shortened, rapid stride can be useful for maintaining a solid pace early on while sparing your legs for harder terrain in later miles.

Stride length and frequency can also be manipulated during drills to prepare you for harder work in interval sessions. For instance, during a warm-up, a series of twenty-second RunningStrides (drills from a standing start, focusing on a constant twenty-second acceleration) helps to develop the neuromuscular pattern for higher-speed running before you have the fitness necessary to maintain those higher speeds. Similarly, when used during a warm-up before a hard interval workout, RunningStrides get your body ready for the hard work that is ahead.

Key Running Workouts

Workouts such as endurance, tempo, and steady state are used by many coaches and athletes. Yet all too often, these words can be confusing and fail to precisely describe the workout in question. In fact, here's a quick test: go ask five of your running friends, "How long/hard should a tempo run be?" You'll certainly get different answers, but what should strike you most is the range of answers. Over the course of my coaching career I have seen durations from one minute to two hours used to describe a single type of workout. At CTS we use these words as well, but with very specific definitions so our coaches can communicate with each other and with their athletes consistently and precisely. Throughout this book, I use the following terminology to describe workouts and their associated intensities.

	RPE	TYPICAL INTERVAL TIME	TOTAL TIME AT INTENSITY	WORK : REST	TYPICAL WORKOUT	FREQUENCY PER WEEK
RecoveryRun (RR)	4 to 5	NA	20–60 min	NA	40–min RR	2–3
EnduranceRun (ER)	5 to 6	NA	30 min– 6+ hours	NA	2–hr ER	2–6
SteadyStateRun (SSR)	7 to 8	20–60 min	30 min– 2 hours	5 to 8:1	2–hr ER with 2 x 30 min SSR, 5–min recovery between intervals	2–4
TempoRun (TR)	8 to 9	8–20 min	30–75 min	2:1	2–hr ER with 3 x 12 min TR, 6–min recovery between intervals	2–3
RunningIntervals (RI)	9 to 10	1–3 min	12–24 min	1:1	90–min ER with 6 x 3 min RI, 3–min recovery between intervals	2–3

TABLE 9.1 The five critical workouts.

[Table 9.1 presents a summary of the five key running workouts and their associated ranges of duration, frequency, interval length, rest between intervals, RPE, and breathing rate. In each individual section below, I will outline how you can tailor these interval combinations depending on your experience level.](#)

RECOVERYRUN (RR)

To be effective, a RecoveryRun needs to be very easy. All you’re trying to do is loosen up your legs and increase circulation and respiration with some mild activity. Your RecoveryRuns should be no more than sixty minutes; the typical duration I prescribe is about forty minutes. Perceived exertion for a RecoveryRun is about a 4 or 5, so it’s not a leisurely walk, but it should be substantially easier than an EnduranceRun. The frequency for RecoveryRuns depends on your training schedule, since this workout needs to be balanced with your harder training sessions, but I frequently have athletes run two or three of them in a week.

ENDURANCERUN (ER)

You’re going to spend much of your running time in the EnduranceRun intensity range. This is the moderate-intensity running time surrounding your focused interval sets, as well as the “forever” intensity for your EnduranceRuns that contain no specific intervals. Perceived exertion for this intensity is 5 or 6 and will naturally vary with uphill and downhill. EnduranceRun durations range from thirty minutes to more than six hours. A typical workout would be a two-hour EnduranceRun. When you are running at this intensity, however, it is

important to slow down when you begin going uphill. It can be easy for your intensity level to creep up into SteadyStateRun or lactate threshold territory, and then you are adding training stress and using energy you may need and want later.

STEADYSTATERUN (SSR)

A SteadyStateRun pushes you to a challenging aerobic pace but keeps you below your lactate threshold intensity and pace. This intensity plays a very important role in developing a stronger aerobic engine because you are maintaining an effort level greater than your normal “forever” pace. SteadyStateRun intensity is very close to your everyday EnduranceRun intensity. Most runners find that they naturally fall into this intensity when doing continuous climbs longer than twenty minutes. SteadyStateRun intervals should be as continuous as possible, with individual intervals ranging from twenty to sixty minutes and total time at intensity for a single workout ranging from thirty minutes to two hours. A typical SteadyStateRun workout might be two thirty-minute SteadyStateRun intervals separated by four minutes of easy recovery. The recovery period is intentionally short as compared to the total duration Compounds of often the interval found in foods in order which to protect facilitate the right intensity, which will be well below against lactate oxidation threshold. or reactions The RPE with free-for a radical SteadyStateRun is 7 to 8, and it will take a bit of extra attention May be consumed to keep in food your forms intensity or as isolated level from creeping up toward TempoRun territory on climbs. chemicals SteadyStateRuns can be done on flat, uphill, or slightly rolling terrain.

RUNNER EXPERIENCE	WORKOUT STRUCTURE	RPE	TOTAL TIME AT INTENSITY
Beginner	1 x 40 minutes hard	7 to 8	40 min
Intermediate	2 x 25 minutes hard with 4 minutes recovery	7 to 8	50 min
Advanced	2 x 30 minutes hard with 4 minutes recovery	7 to 8	60 min
Pro	2 x 45 minutes hard with 4 minutes recovery	7 to 8	90 min

TABLE 9.2 SteadyStateRun examples.

TEMPORUN (TR)

TempoRun intervals are a crucial workout for making you a faster and stronger runner. The pace and intensity for these intervals are strenuous, and you will be running slightly below or at your lactate threshold intensity. These intervals help drive the process of increasing the size and density of mitochondria in your muscles, which improves your ability to process and utilize lactate as fuel. They also can increase plasma volume, giving you greater ability to manage your core temperature and increasing mitochondrial enzyme activity, which helps you produce more energy faster.

The critical component of TempoRuns is the time spent at the particular intensity. Therefore, the design of the interval structure is imperative. The intensity will be too low if there is too much interval volume combined with prolonged interval length and not enough recovery. Intensity might be too high if there is too little volume with a protracted interval length and copious recovery. As a rule of thumb, most trained runners could sustain TempoRun intensity for forty to ninety minutes if you were to do an all-out time trial. That’s a big range, but, in the laboratory and in the field, we see this range play out. Less-experienced runners start out only being able to tolerate forty to fifty minutes at their lactate threshold pace. As they become more trained, they can increase this tolerance to an hour or even longer, even when adjusting for the increase in pace to accommodate their increase in fitness. In other words, training both makes your lactate threshold pace faster and increases your tolerance for longer durations at this intensity.

Accordingly, the total amount of interval time for a TempoRun workout will be between forty and ninety minutes, divided up into intervals of eight to fifteen minutes. Rest between intervals should be half the duration of the interval, meaning a 2:1 recovery ratio, or six minutes of recovery between twelve-minute intervals. TempoRun intervals should be run at an RPE of 8 or 9 with the rate of perceived exertion gradually creeping up throughout the course of each interval and throughout the set. TempoRuns can be done on fl at, uphill, or slightly rolling terrain.

RUNNER EXPERIENCE	WORKOUT STRUCTURE	RPE	TOTAL TIME AT INTENSITY
Beginner	3 x 12 minutes hard with 6 minutes recovery	8 to 9	36 min
Intermediate	4 x 12 minutes hard with 6 minutes recovery	8 to 9	48 min
Advanced	4 x 15 minutes hard with 7 minutes recovery	8 to 9	60 min
Pro	5 x 15 minutes hard with 7 minutes recovery	8 to 9	75 min

TABLE 9.3 TempoRun examples.

RUNNINGINTERVALS (RI)

If there were a workout ultrarunners are allergic to, it would be RunningIntervals. The lung-searing intensity these intervals induce is seemingly incompatible with the long, slow nature of an ultramarathon. Combine that with the sheer discomfort these intervals elicit, and even the most seasoned ultrarunners will avoid these like the plague. Yet, ultradistance athletes can reap important benefits from this type of work. RunningIntervals improve your heart’s stroke volume and maximum cardiac output, and consequently your VO2 max. But the benefits do not end there. Like TempoRuns, RunningIntervals can also increase your blood plasma volume, giving you the ability to tolerate heat better as well as improving muscle capillarization, which helps deliver blood and oxygen to working muscles.

RunningIntervals are performed at an effort coaches call “VO2 max” intensity. The quotes are intentional, as you are only at your true VO2 max for part of the workout. More importantly, and what drives most of the adaptation, is that you are accumulating more than ten total minutes running at an effort that elicits 90 percent of your VO2

max oxygen consumption. Most of the research behind this specific time at intensity (ten minutes at 90 percent of VO2 max) was done by French researcher V  rinoque Billat, who interestingly enough won the famed Sierre-Zinal race in 1982.

RUNNER EXPERIENCE	WORKOUT STRUCTURE	RPE	TOTAL TIME AT INTENSITY
Beginner	4 x 3 minutes hard with 3 minutes recovery	9 to 10	12 min
Intermediate	5 x 3 minutes hard with 3 minutes recovery	9 to 10	15 min
Advanced	6 x 3 minutes hard with 3 minutes recovery	9 to 10	18 min
Pro	5 x 4 minutes hard with 4 minutes recovery	9 to 10	20 min

TABLE 9.4 RunningIntervals examples.

Now that the history lesson is over, let’s get into how to structure a workout to drive this critical time at intensity. In order to drive the critical amount of time (more than ten minutes) at the specific intensity (>90 percent of VO2 max oxygen consumption), I target intervals that are two to four minutes in length and add up to twelve to twentyfour minutes of total work time, with a 1:1 work to recovery ratio between intervals. The RPE for these efforts is between 9 and 10.

Although these intervals can be completed on any terrain or on a treadmill, I recommend doing them uphill if possible. The incline is helpful for increasing the workload and enabling you to reach 90 percent of your VO2 max more consistently. RunningIntervals should never be done downhill, as it will be nearly impossible to achieve >90 percent of your VO2 max in this condition.

Peak and Fade RunningIntervals

Peak and fade RunningIntervals are an advanced training technique I use with many of my athletes who have experience doing interval work. While you don’t need to be an elite runner to partake, you do need to have a moderate amount of training experience under your belt (two to three years running ultras), as well as experience with interval work. If you fit the bill and are up for the challenge, buckle up. Peak and fade intervals will test even the heartiest of athletes. As the name implies, you go as hard as you can during the initial part of the interval (aka the “peak”) and simply fade as you become fatigued further into the interval.

HOW TO DO PEAK AND FADE INTERVALS

Start with the same RunningInterval workout you would normally do (six sets of three minutes hard, three minutes easy, for example). As you begin each interval, take the first fifteen seconds to ramp up to the highest intensity you think you can sustain for sixty to ninety seconds. Hold that intensity for as long as you can. Because the interval is greater than ninety seconds, the pace/intensity will not be sustainable (intentionally), and for the last half of the interval you will gradually slow down while hanging on for dear life. Rinse and repeat for the remainder of the workout. If you are running these on a track or flat ground, you will likely be about 10 percent faster at the beginning of the interval compared to the end. Yet your heart rate will increase throughout the set of intervals.

These intervals can be intimidating. After all, you are intentionally starting each interval at an unsustainable intensity. In reality, they are not all that frightening once you give them a chance. You will never get the pacing perfect, and that's sort of the point. As long as your pace is faster at the beginning than at the end (which you might not be able to measure if you are doing this workout on trails) and they are literally a struggle to complete, you're doing it right.

WHY PEAK AND FADE INTERVALS WORK

Peak and fade intervals can be superior compared to their evenly paced interval counterparts because they drive a higher absolute amount and average of rate of oxygen delivery to the working muscles—and that higher oxygen demand will in turn produce a more robust adaptation. This was elegantly demonstrated in a study by Brent Rønnestad from Inland Norway University of Applied Sciences (Rønnestad et al. 2019). In the study, they had a group of well-trained cross-country skiers complete a workout (five sets of five minutes hard, with three minutes recovery) using a peak and fade model and compared the results with the same workout that was evenly paced. What they found was that the peak and fade intervals produced higher mean and absolute VO₂ levels, indicating that subjects' aerobic systems were stressed more despite the workout construction being identical.

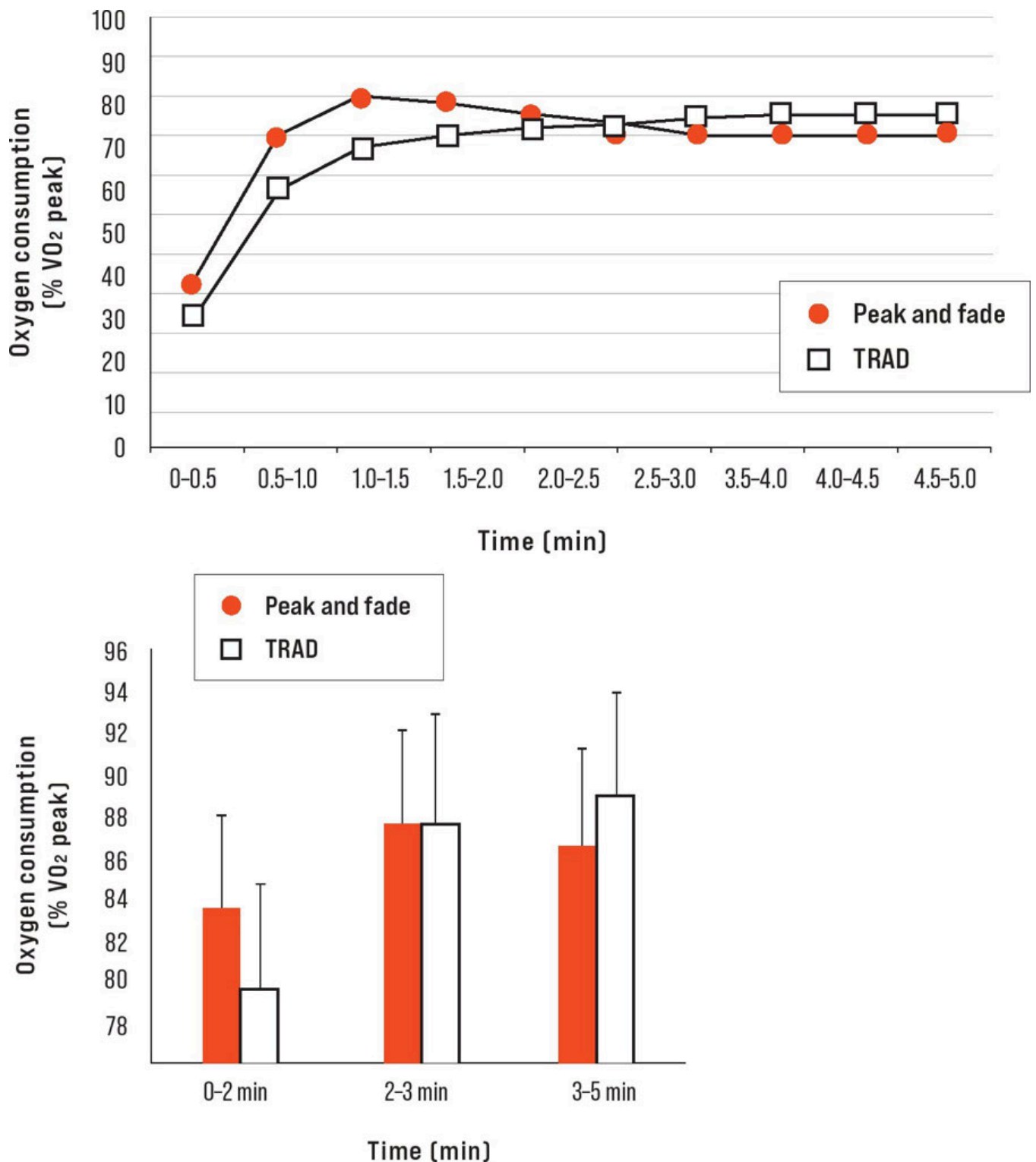


FIGURE 9.1 Oxygen consumption response from traditional, evenly paced intervals (TRAD) and peak and fade intervals. Adapted from Rønnestad et al. 2019.

I also feel there is a toughness component at play with these intervals. Psychologically they are difficult to wrap your head around, as you are not saving anything for the end of the interval or subsequent intervals. However, I feel that once athletes get through this aspect and get comfortable being uncomfortable (even while running slower), it facilitates a new level of grit and determination they can tap into on race day.

		Racing Intensity		Ultramarathon Zone					Maybe if you are an elite			
		% VO ₂ max										
				50	60	70	75	80	85	90		
Workouts		RecoveryRun		EnduranceRun		SteadyStateRun		TempoRun		RunningIntervals		Traditional speed workouts (e.g., 6 x 200 m)
Adaptation	Plasma volume	✓				✓✓		✓✓✓		✓✓✓✓		✓
	Mitochondrial enzyme activity	✓✓		✓		✓✓✓		✓✓✓✓		✓✓		✓
	Lactate threshold	✓✓		✓		✓✓✓		✓✓✓✓		✓✓		✓
	Capillarization	✓				✓✓		✓✓		✓✓✓		✓
	Conversion of Type IIB muscle fibers to Type IIA	✓✓		✓		✓✓✓		✓✓✓		✓✓		✓
	Stroke volume and cardiac output	✓				✓✓		✓✓✓		✓✓✓✓		✓
	VO ₂ max	✓				✓✓		✓✓✓		✓✓✓✓		✓
	Lactate tolerance/ anaerobic capacity									✓		✓✓
	ATP/PCr stores											✓✓
	Muscle glycogen storage	✓✓		✓		✓✓✓✓		✓✓✓		✓✓		
	Neuromuscular power											✓✓

When we as a coaching staff review workout designs, one critical element emerges to determine the efficacy of the intended workout: time at intensity. For each of the aforementioned critical workouts (SteadyStateRun, TempoRun, and RunningInterval), I have gone through the painstaking task of determining how much exposure is needed to elicit a particular adaptation. It's not a task to be taken lightly. After all, you want the work you do to result in improvement. Too much time at intensity and your workout gets shortchanged by the resulting reduction in the intended intensity. Too little time at intensity and there is not enough exposure to elicit the adaptation you want. However, when done correctly these workouts will result in a wide range of physiological adaptations that you can target depending on your goals.

Similar to the energy systems concept introduced in Chapter 2, these adaptations do not turn on and off like light switches. Rather they are tuned up and down like volume knobs depending on the workout. I've summarized the specific physiological responses expected from all of the workouts described in Table 9.5. I have also overlaid common racing intensities seen in the ultramarathon world so that you can relate the workout intensities to what you can expect on race day. Additionally, I've added a separate column to summarize the adaptations from more traditional distance running speed work like 200 m and 400 m repeats.

A Word on Recovery

All this talk about interval RPE, duration, number of intervals, and terrain might make you think, "What about the intensity of the recovery period?" Don't complicate this. Do the work periods hard and the easy periods easy. This means that the recovery between intervals can be easy running or even walking. Remember, the structure behind these workouts is specifically designed to target an amount of time at a physiological intensity which then drives a certain adaptation. Increasing the intensity of the recovery period will normally shift the entire workout to an easier intensity and therefore might force you out of this critical time at intensity you are trying to target.

That last sentence might seem counterintuitive, but it's true. When the intensity of recovery periods is too high, it results in lower intensities for your work intervals, which reduces the effectiveness of the whole workout. Easier rest yields stronger intervals.

The Right Warm-Up for Trail Running

Many run-training programs define precise warm-up and cooldown periods to bookend specific interval work. Typically, and somewhat arbitrarily, the warm-up and cooldown periods are exactly fifteen minutes each. The fifteen-minute number is not magical. The background for this structure is rooted in track and field, where the warm-up and cooldown can be done on roads near the track (and can thus be exactly fifteen minutes in duration). In trail running, I find this approach impractical. I am far more interested in the specificity of the intervals, which should include both the intensity and appropriate environment (either surface or grade characteristics) for the desired adaptation. Many times, this will require the athlete to run for seventeen or nineteen or twenty-two minutes to get to the section of trail that is ideal for the task. Therefore, when I prescribe a workout that includes intervals, I use the EnduranceRun duration to define the total time of the workout, and I intend the intervals to be completed within that time. For instance, a two-hour EnduranceRun with three by ten-minute TempoRuns and five minutes of recovery between intervals means "run a total of two hours, and within those hours complete three ten-minute TempoRun efforts separated by five minutes at RecoveryRun pace." This gives the athlete a flexible amount of time for the warm-up period and allows additional training time to be accumulated on the back side of the interval portion. My only parameters are that the warm-up lasts between ten and thirty minutes and that the cooldown lasts at least ten minutes. Often I also prescribe four to eight RunningStrides of twenty seconds each after the warm-up and before any intervals.

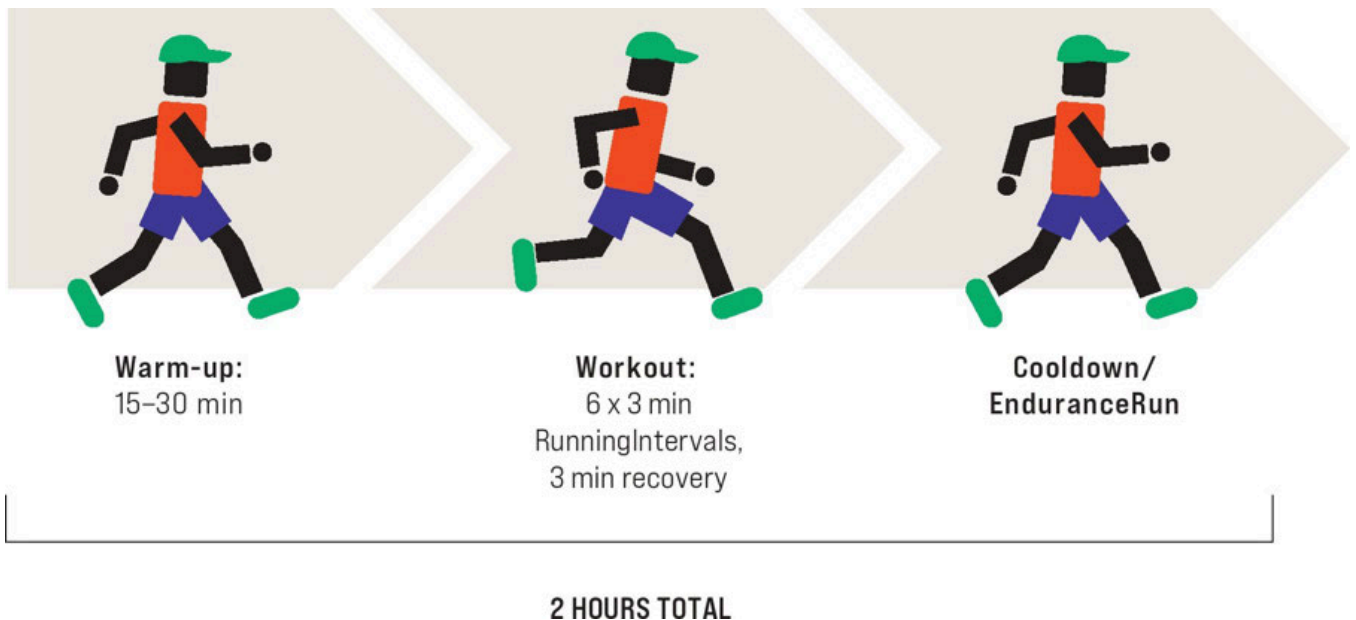


FIGURE 9.2 The right structure for warm-up, workout, and cooldown.

What about Weight Vests or Other Contraptions?

If you have been following the trail and ultrarunning game for any appreciable amount of time, you have undoubtedly seen your fair share of weighted training devices intended to add resistance during running. These come in the form of weighted vests, sleds, carrying rocks in your hands, and other contrived forms of training. Athletes who live in flatter regions often look to these alternative sources of resistance as a means of preparing for a mountainous event.

Weighted vests and other derivatives might have their place in ultramarathon training, but there are some misconceptions about what they are good for and how to use them.

HOW TO USE A WEIGHTED VEST FOR HIKING

There's a lot of hiking in ultramarathons, so from a sport-specificity standpoint it is important to incorporate hiking into training if it is warranted by your goal event. The tricky part, especially for athletes who have limited time to train and who live in flatter areas, is achieving enough cardiovascular stress to create an aerobic training stimulus during hiking and walking. Weighted vests can help deliver the cardiovascular stress of an endurance run while you are hiking.

LOAD UP YOUR RUNNING PACK

Rather than going out and buying a tactical vest with pockets for lead weights or sandbags, load up the pack you intend to use in your event. Fill the hydration bladder with more water than you'll need. Put full bottles in the front storage on your chest. Carry your heavier gear instead of going light. I recommend this because using your pack is more sport-specific—you're distributing the weight as it would be distributed during your event.

DON'T OVERLOAD

When training with a loaded pack, more weight is not necessarily better. Bending forward and slowing cadence is the natural response to shouldering an unnaturally heavy load. That's problematic for a runner because it significantly changes your biomechanics and defeats the purpose of inducing a hike in the first place. The much bigger risk is that it leads to injury.

AIM FOR <10 PERCENT OF BODY WEIGHT OR JUST ENOUGH TO INDUCE A HIKE WHEN YOU WOULD NORMALLY RUN

The primary result of the <10 percent recommendation is that it generally keeps packs under 20 lb as well as ensures more sport-specific biomechanics. For beginners, you might want to start out with ~5 percent of your body weight and then work up/down from there depending on your initial tolerance. Once vests get above 20 percent of body weight (>30 lb for a 150-lb runner), the sport-specific benefits typically decrease due to altered biomechanics, and injury risk increases.

BE CONSERVATIVE GOING DOWNHILL

You're probably going to run and hike downhill wearing your pack during your event, so you should do it in training. However, if you are carrying a pack that is purposely heavier than you're going to use in your event, be cautious about running downhill. As discussed in Chapter 4, downhill running is more stressful from a musculoskeletal standpoint than running in a level or uphill condition. Adding weight to your pack only adds to this stress, which could lead to injury. If convenient, you can load up your pack with extra water for the uphill, dump the water as you head downhill, and repeat the process as you cross streams or return to a refill point.

SKIP PUSHING SLEDS AND PULLING TIRES

A weighted hydration vest is not the only way runners can add resistance, but it's the only one that's really applicable to ultramarathon running. Dragging a tire harnessed to your waist or chest makes it harder to move forward and increases aerobic and muscular workload, but you're not going to drag the tire for miles, and you certainly are not going to see one in any ultramarathon. Pushing or pulling a sled is similar in that the biomechanics are vastly different from ultramarathon running. Both of these activities could be great strength-training activities, but ultrarunners who wish to strength train would be better off doing some heavy squats or reverse lunges. Chapter 11 will have specific recommendations for strength-training activities for ultrarunners.

The Time-Crunched Ultramarathoner

How much time do you really need to train? Most of us are limited by time, with our training somehow crammed into busy lives. Taking an hour at lunch or waking up an hour earlier to get a run in is more often than not the way

most normal people with jobs, friends, and a family fit in their training. It is extremely easy, particularly if you are an aspiring ultrarunner, to conclude that you don't have enough time to train. But I find most athletes overestimate the amount of time truly required to train for an ultramarathon. They tend to linearly expand their marathon training to accommodate longer distances: "If I trained eight hours per week for a marathon, I need to train sixteen for a 50-miler."

On the surface that type of thinking is entirely logical, so I don't blame athletes for approaching the idea of training for an ultramarathon with trepidation. The fact is, though, the relationship between the training required for a marathon and the training required for an ultra is not linear. A 50-miler does not require twice the amount of training as a marathon, nor does a 100-miler take twice as much training as a 50-miler. And even if those volumes of training were ideal, the reality is that most people are limited by available time. Therefore, reality dictates that most people devote a similar amount of time to training volume, irrespective of the distance they are training for. This limitation underscores the need for high-quality structure in your training. After all, if you ain't got much, you better make the most of what you do have.

There is, undoubtedly, a minimum amount of training time required to be successful at an ultramarathon, although it's not the same for everybody or for every distance. I always present this concept in terms of the minimum amount of time you need to be able to devote during your period of highest training volume. This "minimum maximum" sets a reference point for what you can expect to achieve on race day and helps you determine if the distance you have chosen is reasonable. While you do not need to always have this "minimum maximum" amount of training time available, you do need to have it for key weeks during the season:

- 50K and 50 miles: minimum maximum of six hours per week for three weeks, starting six weeks before your goal event
- 100K and 100 miles: minimum maximum of nine hours per week for six weeks, starting nine weeks before your goal event

Outside of this three- or six-week period, you can have a lower volume and be perfectly successful as long as you also do higher-quality training. Although this formula does not guarantee success or maximum performance, not being able to achieve these critical minimum maximums can lead to failure and underperformance.

When setting goals for a season, you need to carefully consider this minimum maximum concept. You need to understand that, according to your goals, you will need to meet these minimum time requirements in key training weeks in order to achieve success. If you can't commit the time, you are less likely to meet your goals. It's that simple. However, if you do have the required time—six hours per week for three weeks, or nine hours per week for six weeks—you have every reason to believe that you can be successful. How successful you are with that time depends entirely upon how effective your training is!

KEY POINTS FOR THE KEY WORKOUTS

There are five key components to a workout: intensity, volume, frequency/repetition, environment, and cadence/stride rate. You can manipulate any or a combination of these components to achieve a desired effect.

The key running workouts used in this book are: RecoveryRun, EnduranceRun, SteadyStateRun, TempoRun, and RunningIntervals.

Each workout type has its own specific structure. Tampering with that structure runs the risk of failing to achieve the desired adaptation.

Interval workouts should be performed with a proper warm-up of ten to thirty minutes that includes easy running, dynamic stretching and strides.

If you are a time-crunched ultrarunner, you can apply the minimum maximum concept to see if you have enough time to train for an ultra: six hours per week for three weeks (50K and 50-mile races) or nine hours per week for six weeks (100K and 100-mile races).

